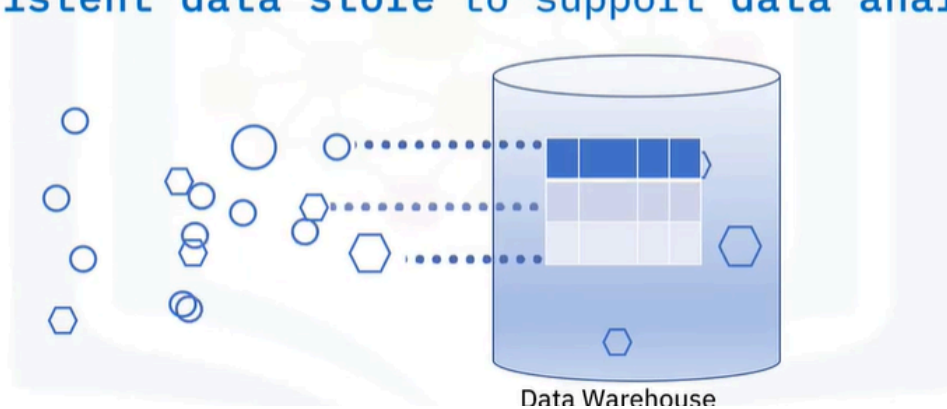


Data Warehouse Overview

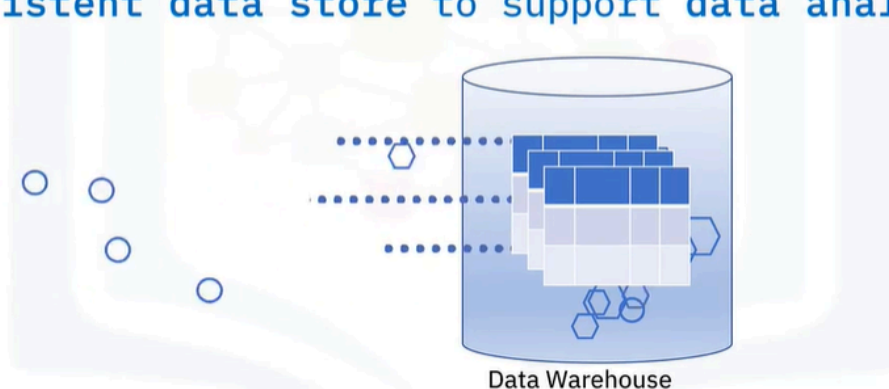
What is a data warehouse?

A data warehouse is a system that aggregates data from one or more sources into a single consistent data store to support data analytics



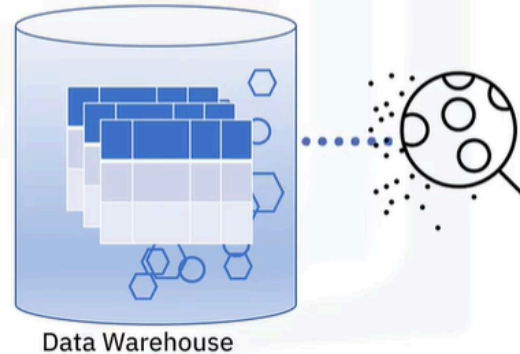
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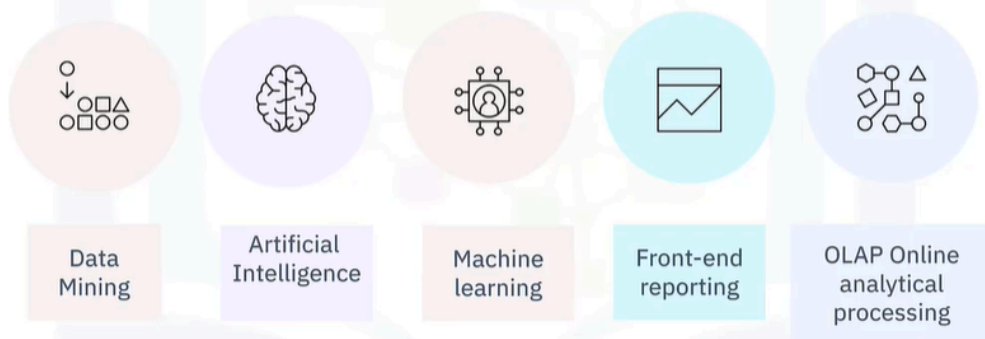
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- A data warehouse is a system that aggregates data from one or more sources into a single, central, consistent data store to support various data analytics requirements. Let's take a closer look at data warehouse analytics.

Data warehouse analytics

Data warehouse systems support:



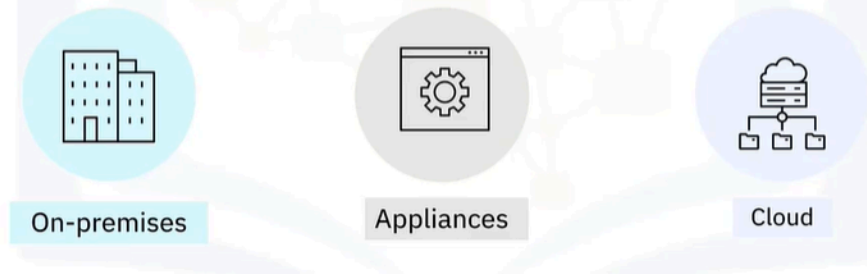
- Data warehouse systems support data mining, including the application of artificial intelligence and machine learning. Data transformation during the ETL process speeds front-end reporting, delivering critical information fast. Data

warehouses enable online analytical processing, known as OLAP, which provides fast, flexible, multidimensional data analysis for business intelligence and decision support applications.

Where are data warehouses hosted?

2010-present

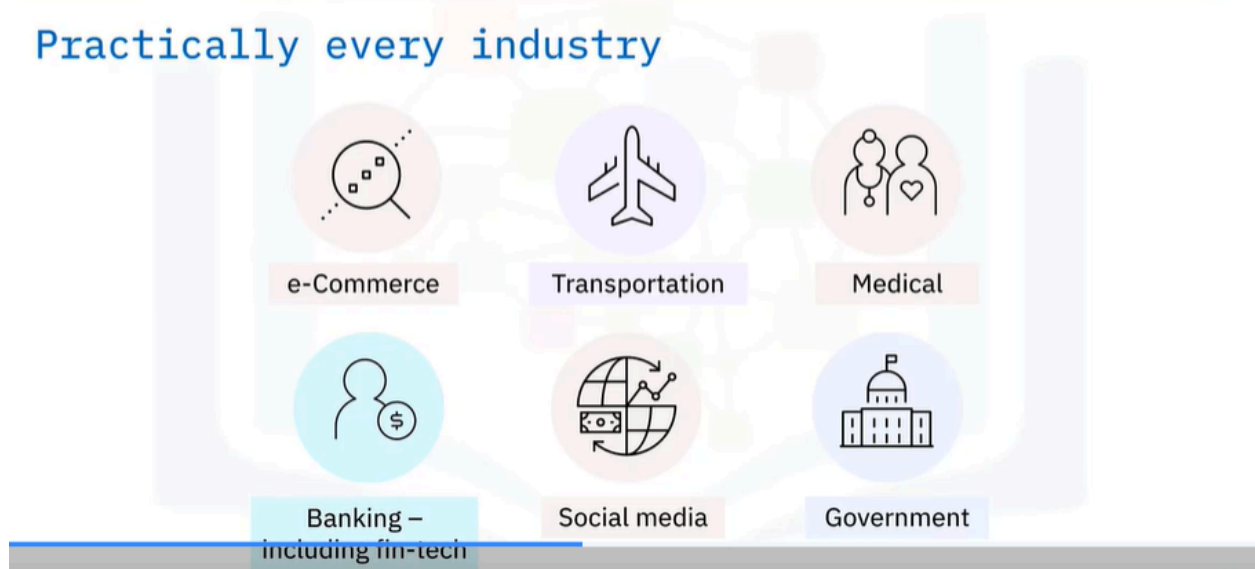
Adoption of Cloud Data Warehouses, providing eliminating hardware purchases, as a scalable service pay-as-you-go service



- Traditionally, data warehouses have been hosted on-premises within enterprise data centers, initially on mainframes and then on Unix, Windows, and Linux systems. Data warehouse appliances emerged with the growth of more extensive data volumes in the 2000s.
- These appliances consisted of a pre-integrated bundle of specialized hardware and optimized data warehousing software that reduced large-scale data warehousing management overhead. In the last decade or so, with exponential amounts of data being generated and stored in the cloud, Cloud Data Warehouses, frequently called CDWs, have gained popularity, where organizations don't purchase hardware or install warehousing software. Instead, organizations access data warehouses as a scalable, pay-as-you-go service.
- Now that you can define what a data warehouse is, identify where data warehouses exist, and understand their basic structure and outputs, let's examine what organizations use data warehouses.

Who uses data warehouses?

Practically every industry



- Data warehouses are a part of almost every industry, including e-commerce, transportation, medical, banking and fin-tech, social media, and governments. But why do these types of organizations use data warehouses?

Retail and e-commerce

- Analyze and report on sales performance
- Create machine learning assisted shopping recommendations



Healthcare



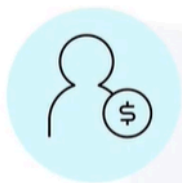
- Apply AI to patient data to assist with diagnosis and treatment

Transportation

- Optimizes routes, travel times, equipment needs & staffing requirements



Banking and fin-tech



- Evaluate risks, detect fraud and cross-sell services

Social media

- Measure customer sentiment and project product sales



Governments

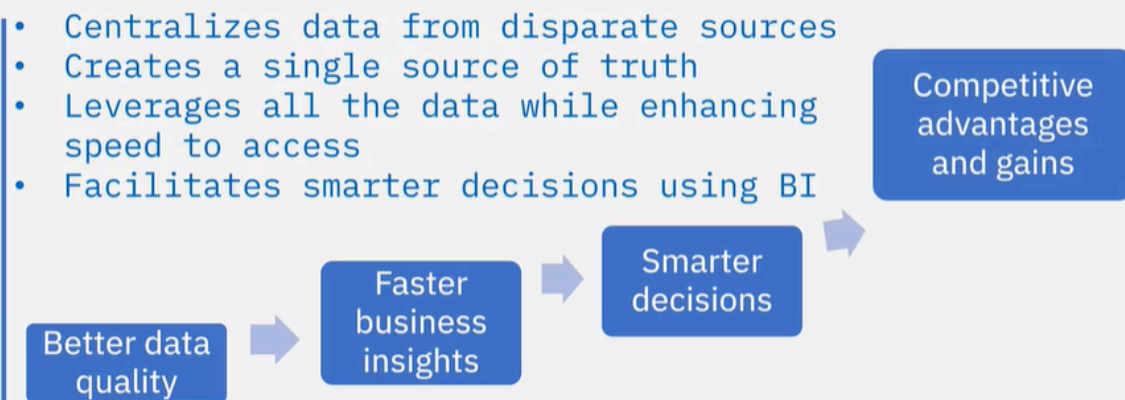


- Analyze and evaluate citizen-focused programs and assist with policy change decisions

- Let's learn more.
- Retail and e-commerce organizations use data warehouses to analyze and report on sales performance. These organizations also apply machine learning assisted shopping that provides shoppers with relevant recommendations that drive additional sales.
- By applying artificial intelligence to patient data, healthcare providers can access the most recent insights and use that information to diagnose and treat their patients with greater accuracy.
- BI capabilities enable transportation providers to optimize routes, travel times, equipment needs, and staffing requirements.

- Financial tech organizations, including banking, apply data analytics to evaluate risks, detect fraud, and cross-sell services.
- Social media organizations need analytic capabilities that can quickly measure ever-changing customer sentiment and project product sales.
- Governments apply business intelligence to analyze and evaluate citizen-focused programs and assist with policy change decisions.

Benefits of a data warehouse



- What are the benefits of a data warehouse? Data warehouses enable organizations to centralize data from disparate data sources, such as transactional systems, operational databases, and flat files. Data integration, removing bad data, eliminating duplicates, and standardizing data create a single source of the truth that results in better data quality for analysis. A single source of truth empowers users to leverage all the company's data and access that data more efficiently. In addition, separating database operations from data analytics generally improves data access performance, leading to faster business insights. Next, large-scale BI functions such as data mining, artificial intelligence, and machine learning tools facilitate smarter decisions by data professionals and business leaders. These capabilities build on each

other to give organizations the means and opportunity to realize competitive advantages and gains.